



**Politecnico
di Torino**

CBD & Aerodynamics CFD LAB Mito CFD analysis

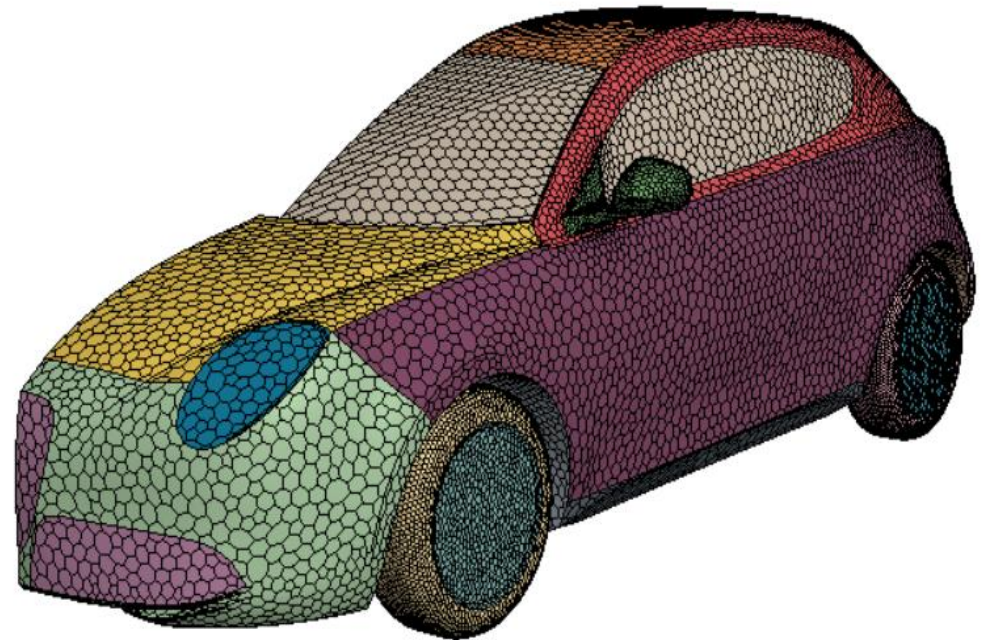
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A.Y. 2025-26

Goals and requested results

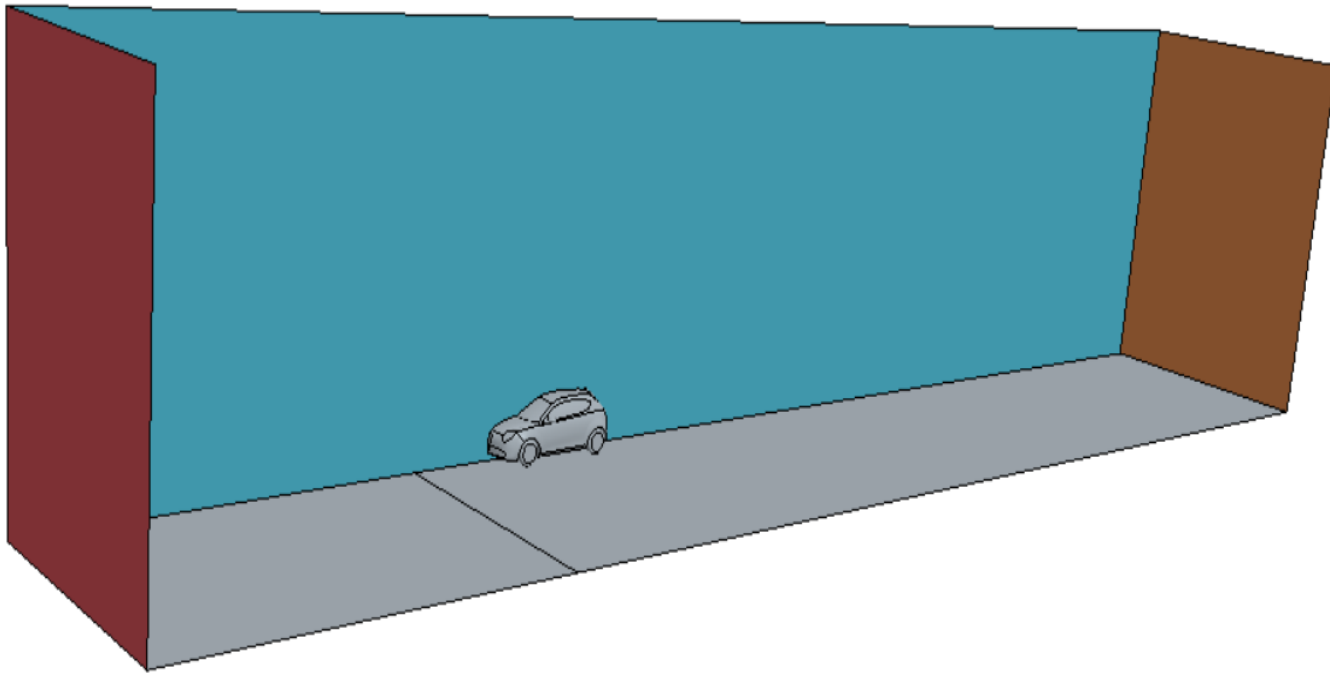
Simulation of aerodynamic performance of an Alfa Romeo Mito using STAR CCM+.

The activity consists in:

- Mesh generation and visualization
- Run the simulation
- Analysis of the result
- Evaluation of C_d and C_l values
- Comparison of the values with experimental results
- Visualization of relevant parameters: isopressures, streamlines, velocity planes, skin friction coefficients, pressure coefficients.



Study case and wind tunnel



- Only half of the 3D vehicle is considered to reduce computational level.
- The car is inserted in a wind tunnel, composed by an inlet, an outlet, a roof and 2 sides.
- The inlet is a velocity inlet. The velocity is equal to 140 km/h (38.89 m/s).
- The outlet is a pressure outlet. The reference pressure is 0 bar.
- The air flow has an initial velocity of 20 m/s.
- Reference air values: density = 1.225 kg/m^3
dynamic viscosity = $1.79 \times 10^{-5} \text{ Pa s}$.

Brief theoretical explanation

Computational Fluid Dynamics (CFD) numerically solves the Navier–Stokes equations, which define conservation laws on a fluid continuum:

- Mass conservation (Continuity)
- Momentum conservation (Navier–Stokes)
- Energy conservation (*if thermal effects are included*)

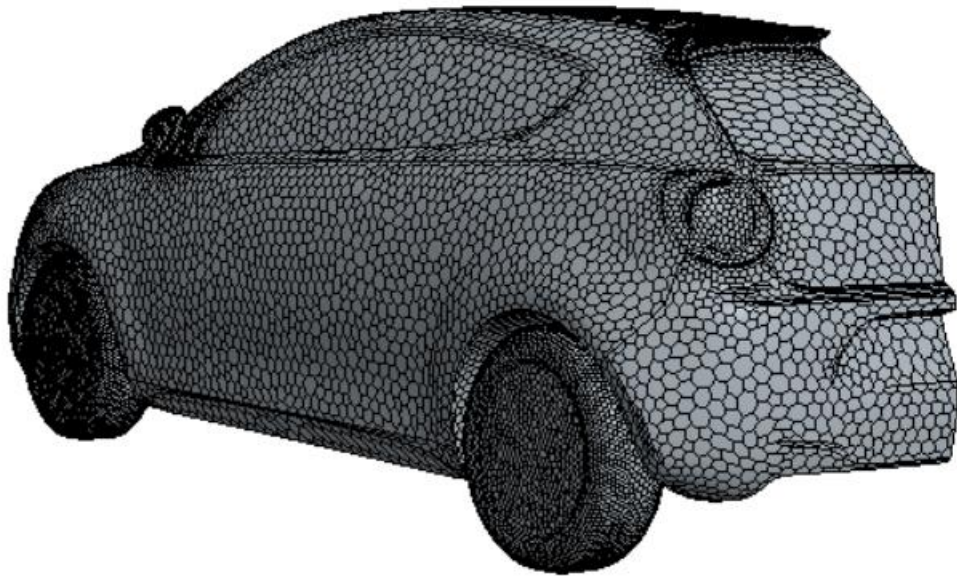
These equations describe how velocity, pressure, and temperature evolve in space and time.

Turbulence Modeling:

- RANS (e.g. k–epsilon, k– ω SST): industrial standard Turbulence fully modeled, low computational cost, robust
- LES: higher accuracy in eddies approximation, higher cost
- DNS: Resolves **all turbulence scales** directly, Used mainly for **fundamental research**

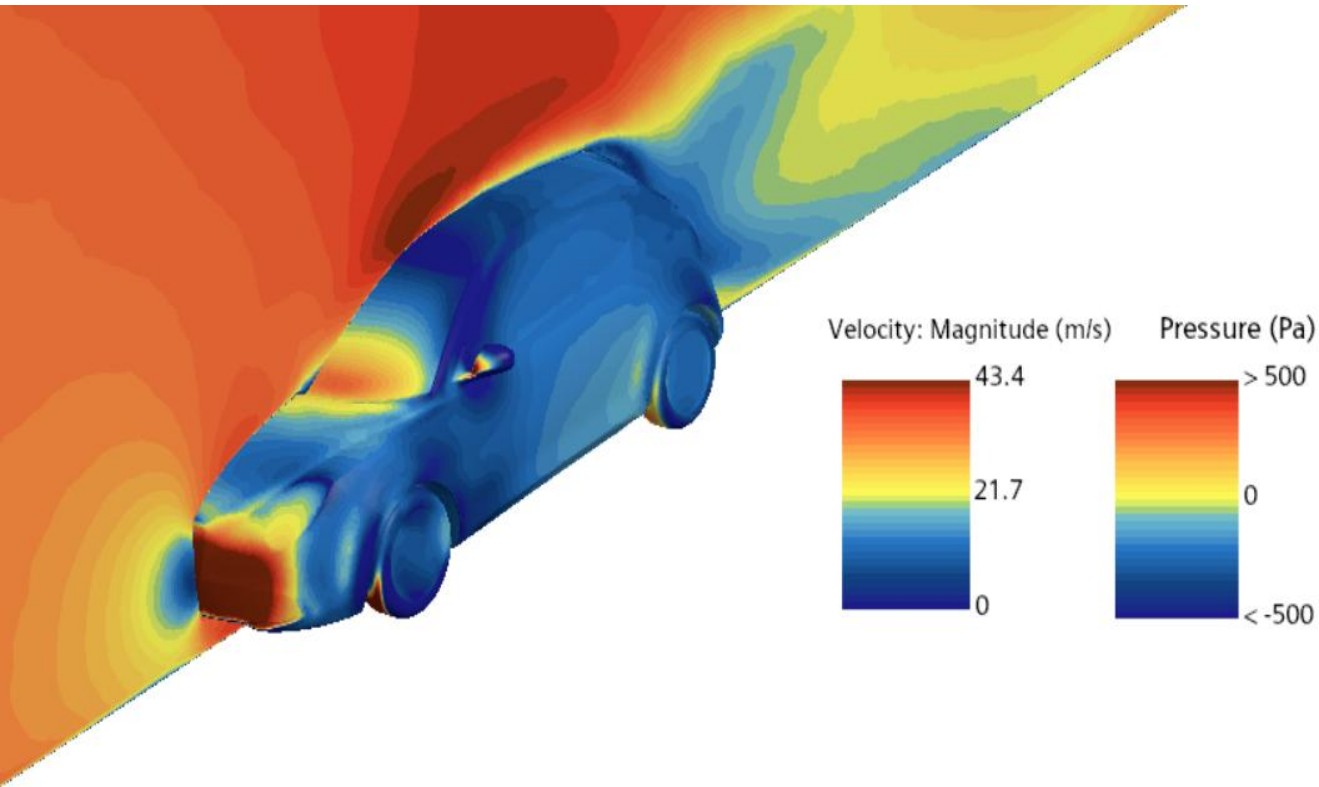
In this simulation, a steady RANS analysis with k-epsilon turbulence model will be adopted.

Mesh



- The mesh is fundamental to perform the simulation: high resolution and quality mesh generally lead to high quality convergence and results .
- The mesh is polyhedral with a base size of 25 mm.
- The wind tunnel is meshed, as well.
- Different reference size are used to increase the mesh density in crucial regions.
- Custom controls are used to thicken the mesh around the rear spoiler, the diffuser, the mirror and the C-pillar.

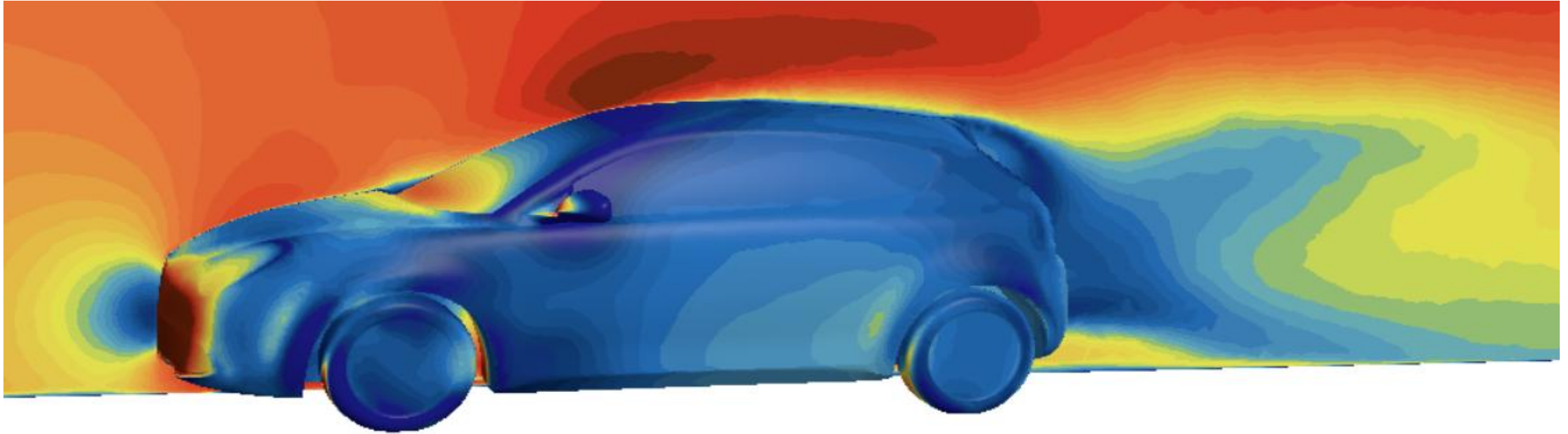
Simulation – velocity and pressure



2 scalar field are displayed: velocity and pressure.

- The flow encounter a stagnation point at the front bumper, where pressure reaches its peak and velocity drops to zero.
- Then the flow accelerates in low pressure zones over the hood, the windshield and the roof. In these regions boundary layer becomes thinner and velocity speed up
- The lower part of the windshield introduce an increase in pressure and a possible flow separation.
- A wake region downstream of the vehicle is characterized by low velocity and low pressure, producing flow separation and turbulences.

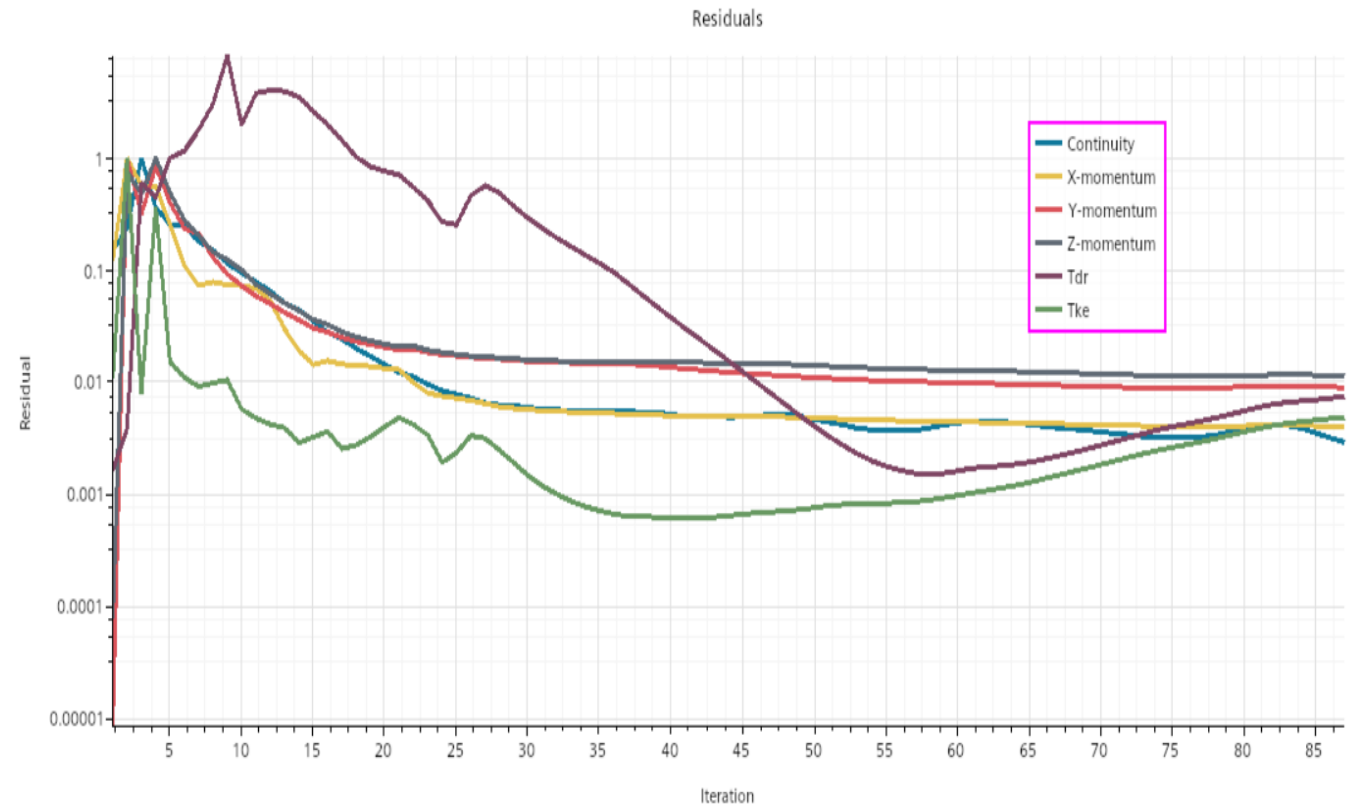
Side view



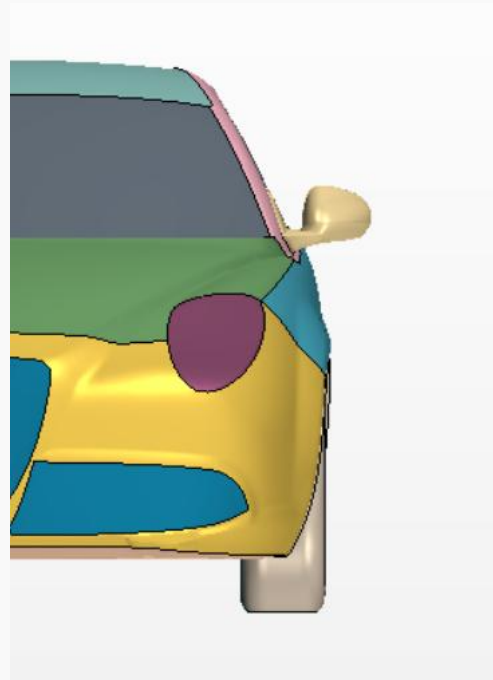
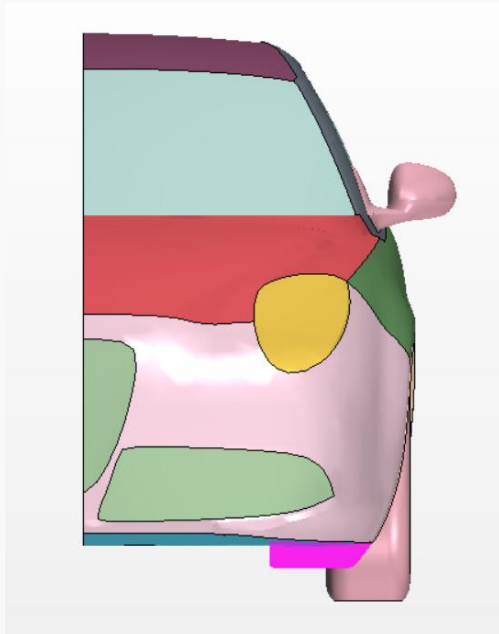
- From the side view the wake behind the car is evident. It is a region of high turbulence with low pressure, that generates a drag increase.
- The boundary layer (denoted by the transition between yellow and blue region on the top of the spoiler) represents the region where the flow separates.
- A region of high velocity is generated on the top of the roof, where the flow speeds up.
- The effect of the mirror is not negligible: it increases drag.

Drag and lift coefficients

- Residuals are indicators of the numerical solution convergence at each iteration (for each equation)
- When simulation is converging, residuals tend to decrease (theoretically to machine precision).
- The simulation provides the following values for C_d and C_l :
- $C_d = 0.28$
- $C_l = -0.05$



Comparison of 2 simulation: effect of a spat



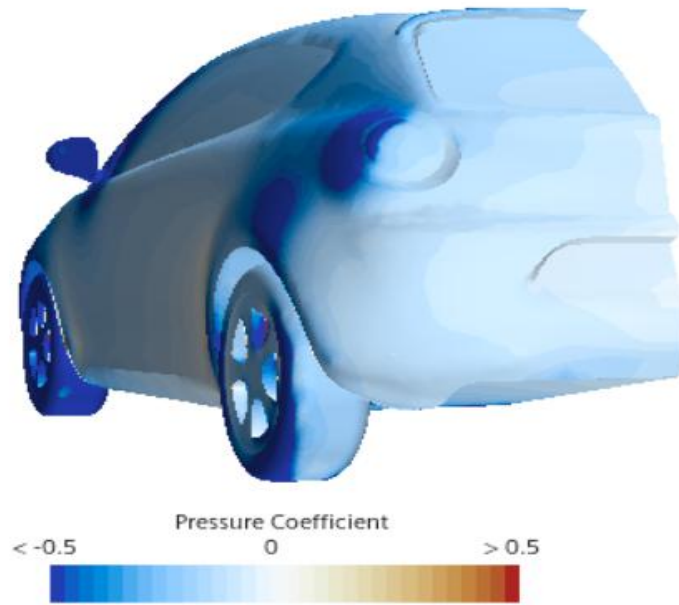
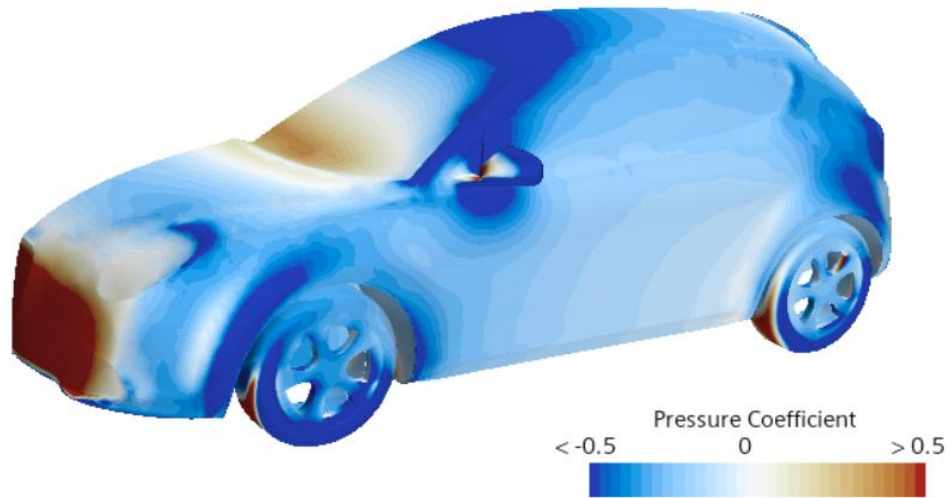
Evaluation of the effect on aerodynamics performance of the introduction of a front wheel spat.

Test conditions:

Velocity = 140 Km/h

RRS ON (Ride Height Reduction System enabled)

EADE ride height (predefined front and rear ride heights used for aerodynamic evaluation)

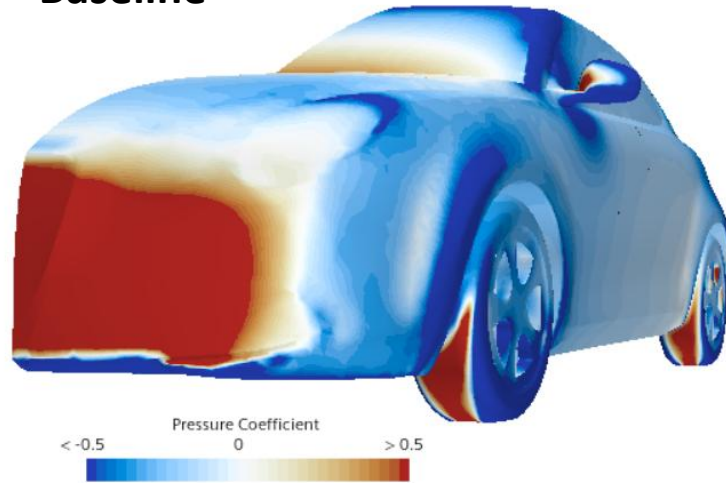


Pressure coefficient

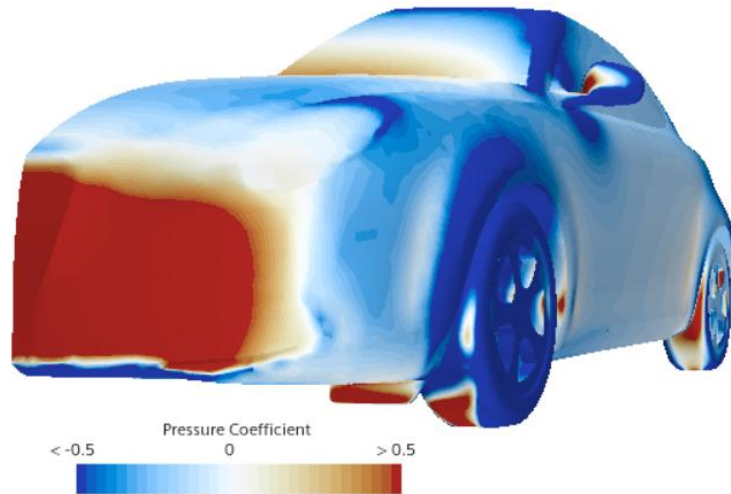
The pressure coefficient influence the drag generation:

- The front bumper exhibits a high-pressure region, that increases drag.
- The lower part of the windshield has the same effect.
- The flow accelerate on the hood and on the roof due to low pressure coefficient. This produce a drag recovery.
- In the rear part, a wide blue region (negative pressure) introduce a strong drag increase.

Baseline



Baseline + spat



Pressure coefficient comparison

The introduction of the spat has 2 main effects:

- It reduces the high-pressure region on the front wheels, as it deflects the flow
- It produces a low pressure around the front wheel arch and the rims.

This generates an acceleration of the flow that can stay more attached to the side of the vehicle, avoiding flow separation.

- The spat influences the pressure on the rear wheel too, by reducing the high-pressure region and, consequently, the drag generation.

Skin friction coefficient

The Surface skin friction coefficient is useful to identify separation lines and wake regions

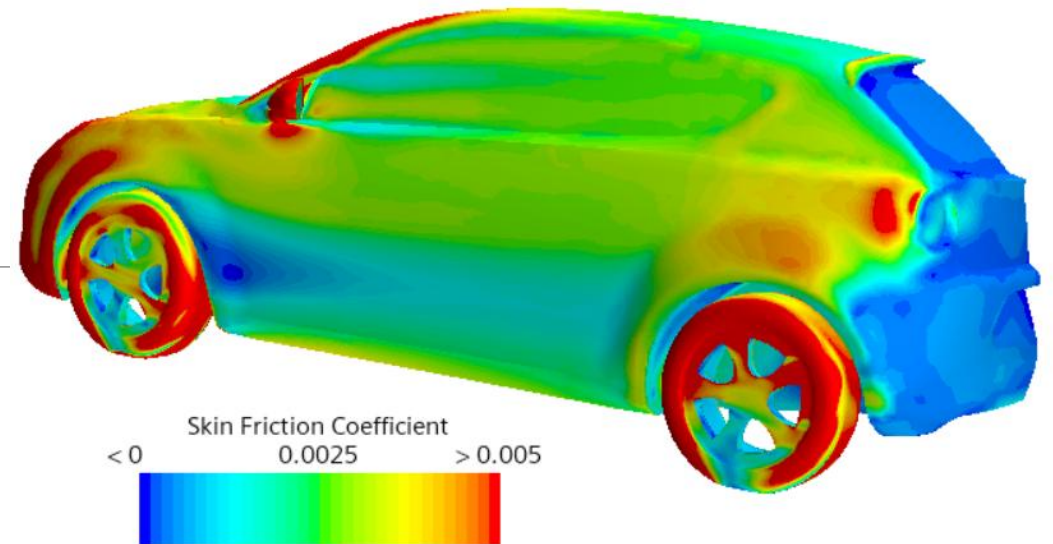
$$C_f = \frac{\tau_w}{1/2 \rho V_\infty^2}$$

A high skin friction → the flow stays attached

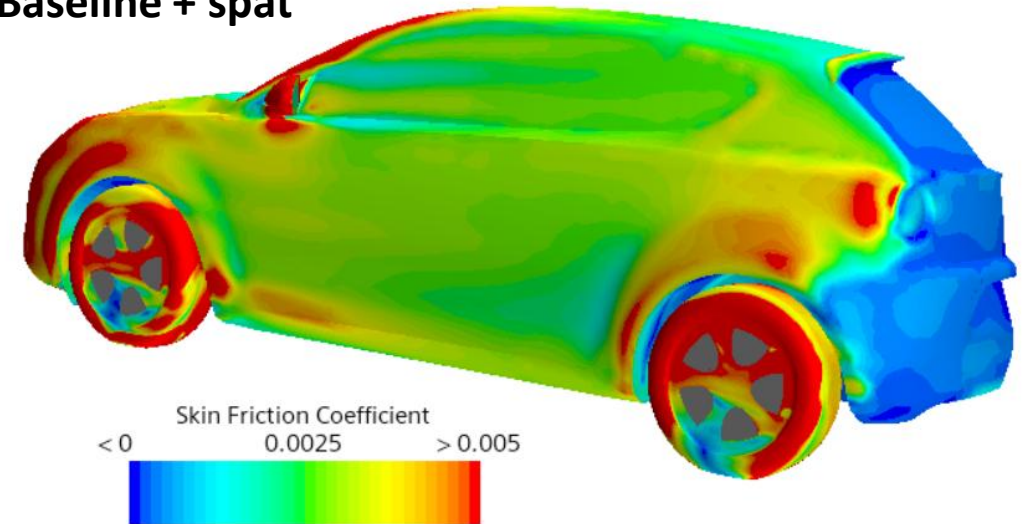
A low skin friction → the flow separates

- The introduction of the spat improves the flow behavior along the side of the vehicle. The skin friction is higher and more uniform, so there is less separation and turbulences.
- Friction is increased also around the wheel arch, where the flow remains more attached.

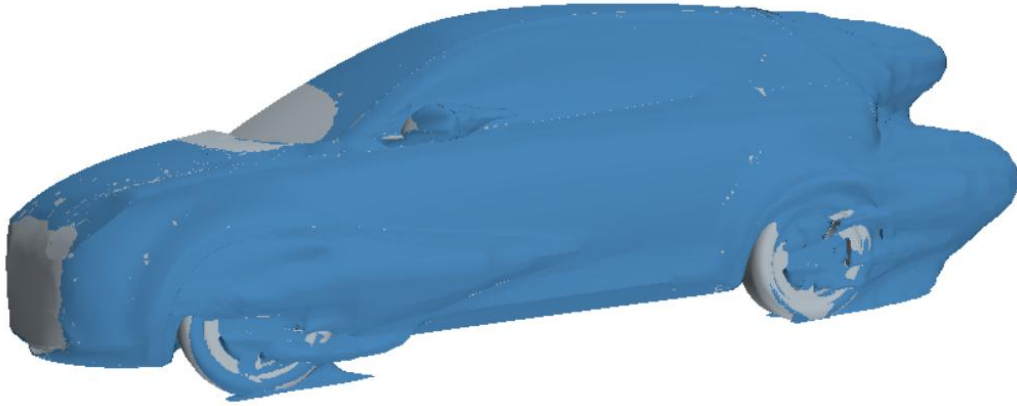
Baseline



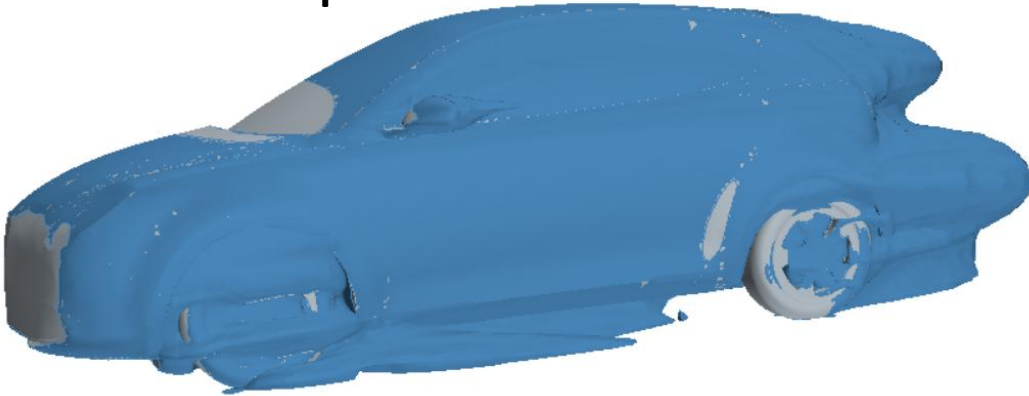
Baseline + spat



Baseline



Baseline + spat



Iso-pressures

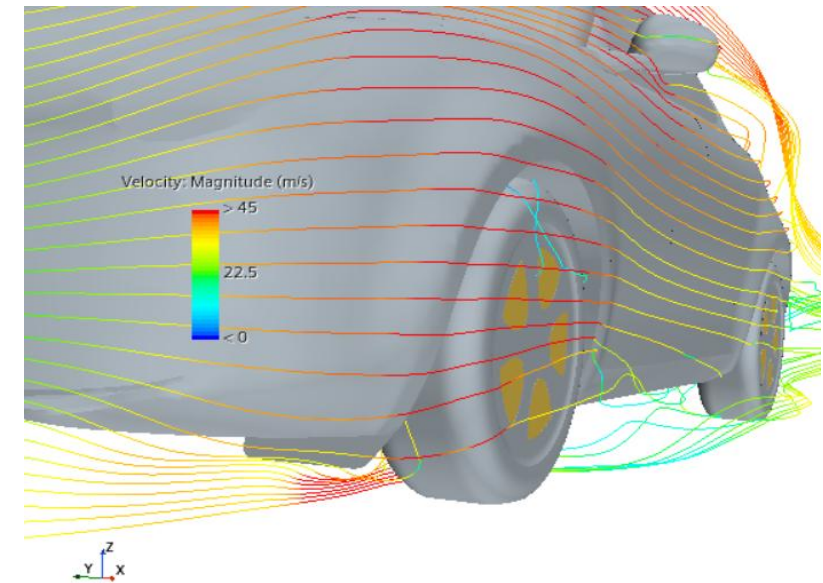
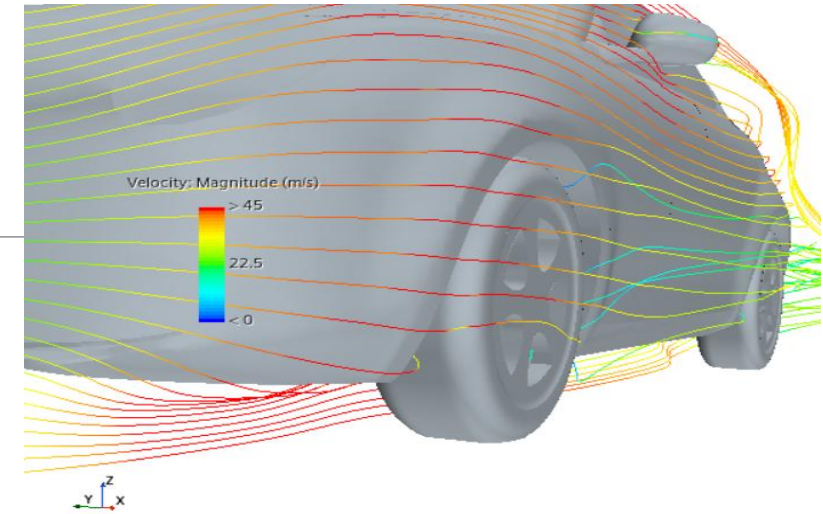
- The iso-pressures are useful to identify the regions of flow separations and the wake.
- The blue region exhibits the regions where $c_p=0$.
- The addition of the spat reduces the flow separation along the side of the vehicle.
- A turbulence region under the floor is evident in the second version. It is important because it reduces the flux that impacts against rear wheels, improving the drag.

Streamlines

A streamline is a path used to visualize flow development. It describes also a scalar quantity, in this case the flow speed.

- The spat deflects the flow, generating important differences.
- In the simulation with the spat the streamlines passing along the side of the vehicle are more uniform and less chaotically distributed. In addition, their speed is higher, providing a better attachment.
- The addition of the spat introduced a flow separation under the floor, denoted by the low-speed streamlines passing under the sills.
- There is also a higher flow recirculation around the wheel and under the wheel arch.

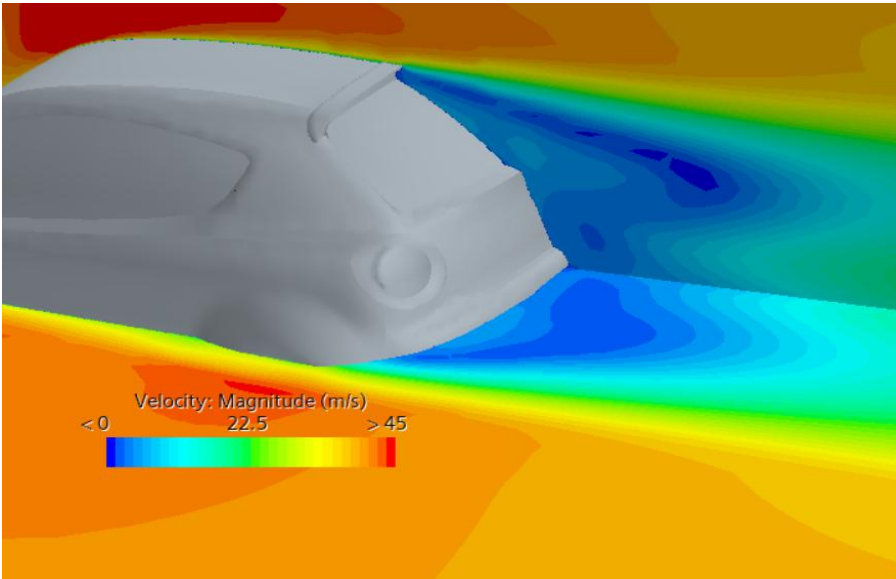
Baseline



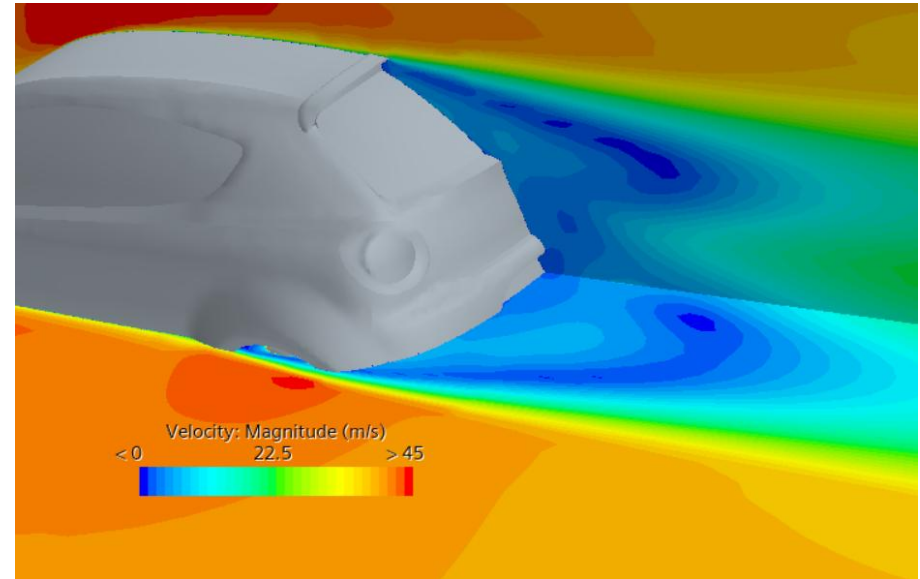
Baseline + spat

Velocity planes

Baseline



Baseline + spat



- Velocity planes show the velocity distribution along two orthogonal plane sections.
- The addition of the spat leads to a thinner boundary layer along the side of the vehicle. Consequently, the flow remains more attached delaying the location of the flow separation point.
- Furthermore, in the wake region the flow is faster, providing a reduction of turbulence and vortices generation.

Cx evolution

There are 3 main contribution to the drag variation:

- The front bumper produces a region of high positive pressure, so the drag coefficient increases.
- The flow speeds up along the windshield, due to a negative pressure. For this reason, there is a drag recovery as the airflow remains attached and laminar.
- The most critical region is the rear section: the steep angle of the rear descent likely causes flow separation. When the flow separates, a low-pressure wake forms behind the object, creating massive pressure drag. The C_x rises exponentially, from 0.10 up to nearly 0.28.

Drag - x development with spat

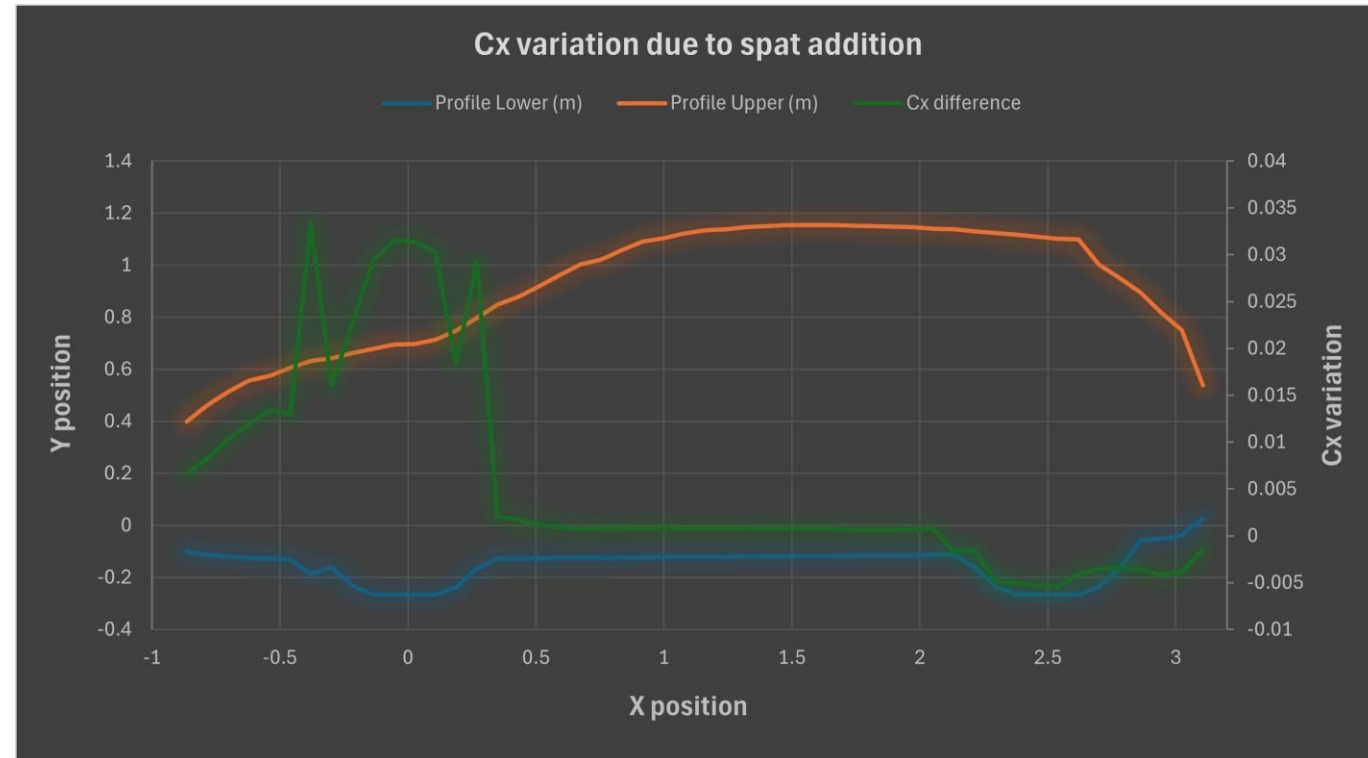


Drag - x development without spat

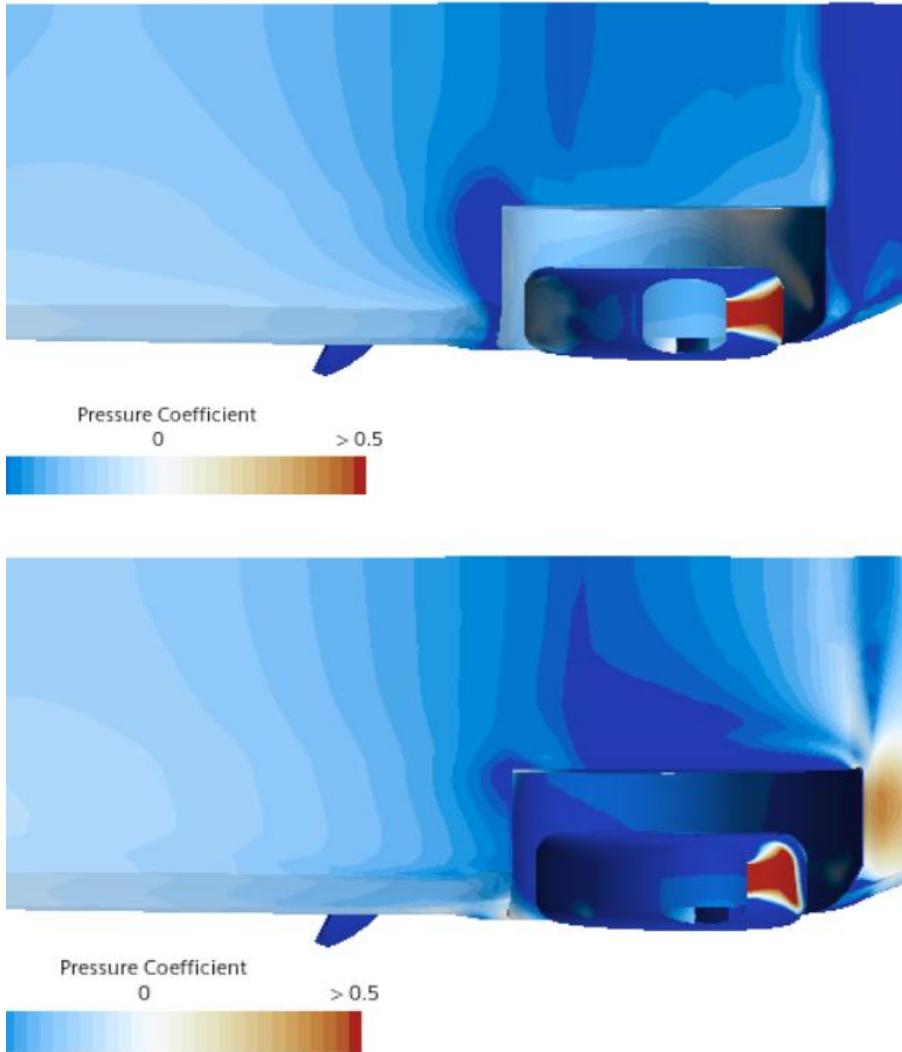


Cx and Cl comparison

- The front wheel spat leads to an overall improvement of the drag coefficient.
- Firstly, the C_x tends to increase due to the high pressure generated by the spat.
- Then, more drag is recovered as the pressure acting on the front wheel is lowered.
- Finally, the flow deflection introduced by the spat reduces the pressure acting on the rear wheel leading to an additional drag recovery.



Comparison of underbody pressure coefficients



- According to the simulations the front wheel spat produces a substantial increase in the C_l value from -0.206 to -0.053.
- This effect is probably overrated. Anyway, it is mainly related to the pressure bubble which is produced in front of the spat.
- This leads to a high-pressure region which weakens the drag along the front part of the floor.

	C_x	C_l
No spat	0.277	-0.206
With spat	0.273	-0.053

Comparison with experimental results

	Simulation	experimental
Cd	0.292	0.290
Cl	-0.043	0.100
ΔC_d	-0.004	-0.006

- The Cd value obtained by the CFD simulation is close to the drag coefficient obtained in the Stellantis climatic wind tunnel, so the simulation is sufficiently accurate.
- The Cd variation due to the spat addition is accurate, too.
- On the contrary, the evaluation of the lift coefficient is far from the experimental values, probably due to the floor of the CAD model, which does not represent underbody components that may produce downforce.



Conclusions

- The CFD analysis using STAR CCM+ proved effective for simulating the aerodynamic performance of the Alfa Romeo Mito.
- The polyhedral mesh allowed to capture critical flow features like stagnation points and wake regions.
- The introduction of the spat induced an improvement of the drag coefficient, decreasing it from 0.277 to 0.273.
- The spat improved performance by deflecting flow away from the front wheels (reducing high-pressure zones) and thinning the boundary layer along the vehicle's side, which delayed flow separation and reduced turbulence.
- The simulation demonstrated high accuracy for the drag coefficient and predicted correctly the drag reduction due to the front-wheel spat.
- The lift coefficient shows significant deviation from the experimental value, mainly due to the simplified CAD model of the underbody.